

Current algebra for chiral gauge theories

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Chiral gauge theories are studied with a special emphasis on the treatment of gauge degrees of freedom so as to obtain a gauge-invariant effective action from which current commutators can be evaluated. It is explicitly shown in a simple example that these commutators are those to be expected in a gauge-invariant theory.

I. INTRODUCTION

It is well known that theories where Weyl fermions interact with gauge fields can be inconsistent due to the existence of anomalies¹ which usually manifest through nonconservation of the fermion current. Quantization of such models recently received renewed interest after the proposal of Faddeev and Shatashvili² concerning the construction of a sensible quantum theory by introducing a new *physical (chiral) field* with a Wess–Zumino action, this leading to a gauge-invariant theory. There has been also progress in the comprehension of how chiral theories can be consistent and unitary although non-gauge-invariant after the works of Jackiw and Rajaraman³ and others⁴ on two-dimensional chiral models.

More recently, it has been shown by Babelon, Schaposnik, and Viallet⁵ that a proper treatment of the gauge degrees of freedom in chiral theories uncovers the presence of the Wess–Zumino action: *the group of gauge transformations acquires the status of a physical field and the anomaly is absorbed*. This phenomenon, which also happens with the Liouville action in the quantization of strings, as first shown by Polyakov,⁶ causes the Wess–Zumino action to naturally emerge in the process of quantization without *ad hoc* introduction of additional fields. On the contrary, a group-valued field, which at the classical level was “lost” due to gauge invariance, naturally reappears after quantization. Similar ideas were independently developed by Harada and Tsutsui.⁷

Following this approach in the present work we give an analysis leading to the current algebra of chiral gauge theories. Although the current commutators we explicitly evaluate correspond to a simple (two-dimensional) model, the scheme we develop, based in the definition of an effective gauge-invariant action from which current–current correlation functions can be computed, is applicable to realistic four-dimensional theories. Some of the results in this paper have been already discussed in previous works.^{3–7} We think, however, that it is worthwhile to present them in light of the framework established in Ref. 5 with the aim of clarifying some obscure points concerning quantization of chiral gauge theories, in particular, in connection with current algebra.

II. THE GENERAL TREATMENT

The developments in Ref. 5 start from the observation that, when quantizing a gauge theory in the path-integral approach, the generating functional should be considered as an integral over the whole space of connections rather than as an integral over the orbit space. This distinction causes no harm when Dirac fermions are present (no axial couplings) since, as first shown by Faddeev and Popov⁸ for pure Yang–Mills theories, an integration over the gauge group factorizes. On the contrary, when Weyl fermions are present, the Faddeev–Popov procedure has to be revised and a trivial factorization is no more valid. Instead, a Fujikawa Jacobian⁹ arises leading to an effective theory which contains the gauge group as a physical field and which is, remarkably, gauge invariant.

Let us start by briefly describing this facts by using the original Faddeev–Popov approach (a more formal presentation can be found in Ref. 5). The generating functional for a gauge theory with chiral (left handed for definiteness) fermions is

$$Z = \int \mathcal{D}A_\mu \mathcal{D}\bar{\Psi} \mathcal{D}\Psi e^{iS[A, \bar{\Psi}, \Psi]} \quad (1)$$

with

$$S = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi} D(A)\Psi, \quad (2)$$

$$D(A) \equiv (i\partial + A)(1 - \gamma_5)/2, \quad (3)$$

where the left-handed projector ensures that fermions have only one chirality. As stressed in Ref. 10, Z should be considered as an integral over the whole space of connections A_μ rather than as an integral over the orbit space. Usually, in the gauge fixing procedure, one passes from the former to the latter by factorizing a gauge-group integration; however, when Weyl fermions are present the standard procedure has to be reconsidered. Indeed, let us write

$$1 = \Delta_{\text{FP}}(A) \int \mathcal{D}g \delta[F[A^g]], \quad (4)$$

where the Faddeev–Popov determinant $\Delta_{\text{FP}}(A)$ is obviously gauge invariant,

$$\Delta_{\text{FP}}[A] = \Delta_{\text{FP}}[A^g] \quad (5)$$

with

$$A_\mu^g = g^{-1}A_\mu g + (1/i)g^{-1}\partial_\mu g \quad (6)$$

and $\mathcal{D}g$ an appropriate measure over the gauge group. After insertion of identity (4) in (1) we get

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$$Z = \int \mathcal{D}A_\mu \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \mathcal{D}g e^{iS[A, \bar{\Psi}, \Psi]} \delta[F[A^g]] \Delta_{\text{FP}}(A). \quad (7)$$

In trying to factorize the group integration one usually exploits gauge invariance of the action S , of Δ_{FP} and of $\mathcal{D}g$. Also

$$\mathcal{D}A_\mu = \mathcal{D}A_\mu^g. \quad (8)$$

Now, when writing a relation of the kind (8) for the Weyl fermion measure, one has to include a Fujikawa Jacobian⁹ in the form

$$\mathcal{D}\bar{\Psi} \mathcal{D}\Psi = J(g, A) \mathcal{D}\bar{\Psi}^g \mathcal{D}\Psi^g, \quad (9)$$

since it is not possible to define in this case a gauge invariant measure (see below).^{11,12} We have explicitly indicated the A_μ dependence of the Jacobian so as to stress its appearance due to the regularization procedure.

After relabeling variables one finally has

$$Z = \int \mathcal{D}A_\mu \delta[F[A]] \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \mathcal{D}g \times J(g, A^{g^{-1}}) e^{iS[A, \bar{\Psi}, \Psi]} \Delta_{\text{FP}}(A). \quad (10)$$

As explained above, the presence of the Jacobian prevents the factorization of the gauge group integration and so *the g field has acquired the status of a physical field*. Also, as it was shown in Ref. 5 the effective action defined by integration over fermions and the g field,

$$e^{iS_{\text{eff}}[A]} \equiv Z_{\Psi, \bar{\Psi}, g} = \int \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \mathcal{D}g \times \exp\left(i \int \bar{\Psi} D(A) \Psi dx\right) J(g, A) \quad (11)$$

is gauge invariant, i.e., it depends on the gauge equivalence class of A_μ and not on the particular choice of a representative on the orbit. This independence on the gauge condition choice then translates into BRS invariance.⁵

Let us now discuss with more detail the evaluation of the Jacobian defined by relation (9). The natural measure for fermions, $\mathcal{D}\bar{\Psi} \mathcal{D}\Psi \exp(i \int \bar{\Psi} D(A) \Psi dx)$, once integrated defines the fermion determinant. Hence one gets from (9):

$$J(g, A) = \det D(A) / \det D(A^g). \quad (12)$$

Now, as it is well known, the definition of Weyl fermion determinants is problematic since the corresponding Dirac operator maps negative chirality spinors into positive chirality ones and, consequently, it does not have a well-defined eigenvalue problem.^{9,11,12} Precisely this problem is at the root of the anomalous behavior of chiral gauge theories. To handle this, one can define¹² an operator $\hat{D}(A)$ acting on Dirac fermions,

$$\hat{D}(A) = D(A) + i\partial(1 + \gamma_5)/2 = i\partial + A(1 - \gamma_5)/2, \quad (13)$$

which then leads to a well-defined eigenvalue problem. [As explained in Ref. 12 the doubling in the number of degrees of freedom implied by (13) affects only the overall normalization of the fermion integral since the positive chirality pieces do not couple to the gauge field.] One then defines

$$\det D(A) \equiv \det \hat{D}(A)|_{\text{Reg}}, \quad (14)$$

with the rhs appropriately regularized since the product of

eigenvalues of the Dirac operator (13) grows without bound. The crucial point in this scheme is that the $\hat{D}(A)$'s eigenvalues are not gauge invariant, this being at the origin of the nontriviality of J . Concerning the regularization prescription, the guiding precept is usually gauge invariance. In the present case, however, once definition (14) has been adopted there is no more gauge invariance principle to invoke and, if one uses, for example, the heat-kernel regularization technique, more general regulators than $\hat{D}(A)$ can be used. As we shall see below, this fact can be exploited to render the resulting theory not only gauge invariant but also unitary.

It is important to note that the Jacobian (12) can be identified with a one cocycle¹³ $w_1(A; g)$:

$$J(g, A) = e^{-2\pi i w_1(A; g)}, \quad (15)$$

which satisfies the condition

$$w_1(A^h; h^{-1}g) = w_1(g, A) - w_1(h, A), \quad (16)$$

this being at the origin of the explicit gauge invariance in the generating functional (11).

The Jacobian can be computed by integration of the axial anomaly. Indeed, consider an infinitesimal transformation

$$g = 1 + i\delta\theta, \quad (17)$$

whose corresponding Jacobian is⁹

$$J(\delta\theta) = \exp\left(i \text{tr} \int \mathcal{A}(A) \delta\theta dx\right) \quad (18)$$

which $\mathcal{A}$ the consistent anomaly,

$$\langle D_\mu j^\mu \rangle_{\bar{\Psi}, \Psi} = \mathcal{A}(A) = \frac{1}{i} \frac{\delta \log}{\delta \theta} J(\delta\theta, A). \quad (19)$$

The finite transformation Jacobian is gotten just by iteration of infinitesimal transformations. The simplest way to do it is to introduce a parameter t , $0 \leq t \leq 1$, to build up the finite transformation from the infinitesimal one. The answer is¹⁴

$$J(g, A) = \frac{\det D(A)}{\det D(A^g)} = \exp\left(i \text{tr} \int \mathcal{A}(A^{g(t)}) \theta dt\right), \quad (20)$$

where

$$g(t) = e^{it\theta}. \quad (21)$$

Coming back to the effective action (11), note that it is analogous to that proposed by Faddeev and Shatashvili.² In the present formulation it is important to note that no chiral field has to be introduced *ad hoc* to recover gauge invariance. It is the gauge group which plays a physical rôle solving the anomaly problem. Of course $g(x)$ became physical after quantization since it trivially decouples the classical equations of motion.

From the effective action (11) one can compute v.e.v.'s of product of currents just by differentiation. For example the current-current correlation function is given by

$$G_{\mu\nu}(x, y) = \langle j_\mu(x), j_\nu(y) \rangle = \frac{\delta^2 S_{\text{eff}}[A]}{\delta A^\mu(x) \delta A^\nu(y)} \quad (22)$$

with $j_\mu = e\bar{\Psi}\gamma_\mu\Psi$.

Current commutators can be evaluated from (22) by

means of the Bjorken–Johnson–Low method.¹⁵ We shall now show, by studying a simple example, how these commutators are those expected in a gauge invariant theory.

III. A TWO-DIMENSIONAL EXAMPLE

We shall consider the chiral Schwinger model³ (two-dimensional quantum electrodynamics with Weyl fermions). In order to obtain S_{eff} we have to compute the Jacobian from (12) or (20). Since the model is two dimensional, each determinant in J (and not only its ratio) can be computed exactly. As stated above, there is a regularization freedom associated to the noncovariance of the operator $\hat{D}(A)$ used to define the Weyl-fermion determinant. Following Ref. 14, the fermion determinant can be written as

$$\det D(A) \equiv \det \hat{D}(A) = \mathcal{N} \exp \left(-i \int d^2x \int_0^1 dt \operatorname{tr} \gamma_5 (\phi - \eta) \right)_{\text{Reg}} \quad (23)$$

with

$$A_\mu = -(1/e)(\epsilon_{\mu\nu} \partial^\nu \phi - \partial_\mu \eta), \quad (24)$$

and Reg meaning some regularization prescription. In two space-time dimensions determinants can be evaluated exactly since a chiral change transforms the problem to a free one (see Refs. 14 or 16). In the usual Schwinger model (Dirac fermions) this fact naturally leads to the following regularization:

$$\det \hat{D} = \lim_{M^2 \rightarrow \infty} \mathcal{N} \times \exp \left(-i \int d^2x \int_0^1 dt \operatorname{tr} e^{D^2/M^2} \gamma_5 (\phi - \eta) \right), \quad (25)$$

with

$$\hat{D}t = e^{(\gamma_5 \phi - i\eta)t} (i\hat{D} + A) e^{(\gamma_5 \phi + i\eta)t}. \quad (26)$$

This choice ensures gauge invariance (see Ref. 14 for a detailed discussion). When Weyl fermions are present, a more general $\hat{D}t$ is acceptable since, as we stated before, $\hat{D}(A)$ is not gauge covariant. Following Refs. 4 we shall choose, instead of (26)

$$\hat{D}t = e^{(i/2)(1+\gamma_5)(\eta-\phi)t} \times [\hat{D}(A) + (a/2)A(1+\gamma_5)] e^{-(i/2)(1+\gamma_5)(\eta-\phi)t}, \quad (27)$$

with a a parameter to be determined demanding unitarity and consistency.³ One then finds for the determinant (23),

$$\det D(A) = \exp \left(\frac{i}{8\pi} \int d^2x [(1+a)\phi \square \phi + (1-a)\eta \square \eta - 2\eta \square \phi] \right). \quad (28)$$

Now, the effective action (11) can be written in the form

$$e^{iS_{\text{eff}}[A]} = \int \mathcal{D}g J(A_\mu^{g^{-1}}, g) \det D(A) = \int \mathcal{D}g \det D(A^{g^{-1}}) \quad (29)$$

with

$$g = e^{i\theta}.$$

Using (28) with $\eta \rightarrow \eta - \theta$ and explicitly performing the Gaussian θ integration we finally get

$$S_{\text{eff}}(A) = \frac{-i}{8\pi} \frac{a^2}{1-a} \int \phi \square \phi d^2x, \quad (30)$$

which coincides with the normal Schwinger model result for $a = 2$. The important point is that (30) is gauge invariant for arbitrary a . As we shall see, the current commutators are those expected in a gauge-invariant theory provided $a > 1$ so that the Schwinger term has a coefficient with the correct sign ensuring positivity.¹⁷ This condition for a can be also derived by noting that the effective action (30) plus the $F_{\mu\nu}^2$ term corresponds to a “photon” with a mass $-(1/4\pi)(e^2 a^2/1-a)$ and hence for $a < 1$ the model would produce tachyons.³ One can explicitly construct the current commutators from (30) and (22). One gets for $G_{\mu\nu}(x, y)$:

$$G_{\mu\nu}(x, y) = -(ie^2/4\pi)[a^2/(1-a)] \times (\delta_{\mu\nu} \partial_x^\alpha \square_{xy}^{-1} \partial_{y\alpha} + \partial_{\mu x} \partial_{y\gamma} \square_{xy}^{-1}) \quad (31)$$

with

$$\square_{xy}^{-1} = (1/2\pi) \log|x-y|. \quad (32)$$

Following the Bjorken–Johnson–Low method¹⁵ one writes

$$\langle [j_\mu(x), j_\nu(y)]_{et} \rangle = \lim_{\epsilon \rightarrow 0} [G_{\mu\nu}(x_1, x_0; y_1, x_0 + \epsilon) - G_{\mu\nu}(x_1, x_0; y_1, x_0 - \epsilon)], \quad (33)$$

finally getting

$$[j_0, j_1]_{et} = (-ie^2/4\pi)[a^2/(a-1)]\delta'(x-y), \quad (34)$$

$$[j_0, j_0]_{et} = 0, \quad (35)$$

which are the relations expected in a gauge-invariant theory. Note that with our conventions the Schwinger term coefficient has to have a negative sign¹⁷ this forcing $a > 1$.

To conclude, both the general analysis leading to the effective action (11) and the results for the current commutators in the simple model indicate that gauge invariance is maintained at the quantum level in chiral models if the gauge degrees of freedom are properly treated. No extra matter fields (as can be envisaged following Ref. 18) or chiral fields (as advocated in Ref. 2) have to be *ad hoc* introduced. It is the gauge group which becomes physical, solving the problem of gauge anomalies. It is interesting to note that, although gauge invariance is maintained, only using the regularization freedom (at least in the two-dimensional example) ensures a consistent unitary theory. We shall return to this point thoroughly in a future work.

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¹See, for example, R. Jackiw, in *Current Algebra and Anomalies*, edited by S. Treiman, R. Jackiw, B. Zumino, and E. Witten (World Scientific, Singapore, 1985), and references therein.

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